# MOTOR VIBRATION SIGNAL ANALYSIS

## ADXL345 + ESP32 Experimental Measurement

| Item | Value |
|---|---|
| Dataset | dataload1\processed\adxl345_5sec_calibrated.csv |
| Date | 2026-08-20 22:11 |
| Sampling Frequency | 1559.000 Hz |
| Samples | 4.993e+04 |
| Duration | 32.0269 s |

# 1. Objective

The objective of this script is to analyze experimentally measured motor vibration data, extract time-domain and frequency-domain features, visualize the measured acceleration signals, and prepare a professional engineering report. The analysis is descriptive and comparative; it does not classify the motor condition or automatically diagnose faults.

# 2. Experimental Data

| Parameter | Value |
|---|---|
| Input file | dataload1\processed\adxl345_5sec_calibrated.csv |
| Detected X column | X_g |
| Detected Y column | Y_g |
| Detected Z column | Z_g |
| Acceleration units | g |
| Sampling frequency source | Configured FS parameter in analysis.py |
| Sampling frequency | 1559.000 Hz |
| Number of samples | 4.993e+04 |
| Duration | 32.026940 s |
| Nyquist frequency | 779.500 Hz |
| Frequency resolution | 0.031224 Hz/bin |

# 3. Signal Processing

The input data are already calibrated acceleration values in g, so this script does not perform sensor calibration. The calibrated signals are preserved. For vibration feature extraction and FFT, the remaining DC component is removed using x_ac = x - mean(x), y_ac = y - mean(y), and z_ac = z - mean(z). The resultant signal is R = sqrt(X^2 + Y^2 + Z^2), representing the magnitude of the three-axis acceleration vector; its AC vibration component is obtained by removing its mean. FFT spectra are calculated from DC-removed signals using a Hann window to reduce spectral leakage. Frequency resolution is FS/N.

# 4. Time-Domain Analysis

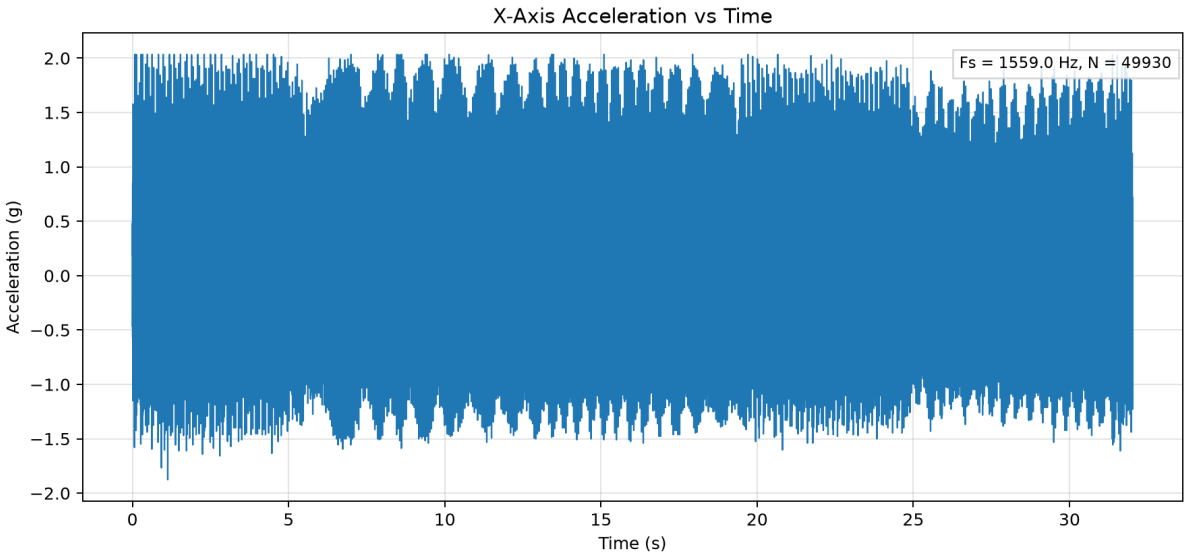



Figure 1. X-axis calibrated acceleration versus time.

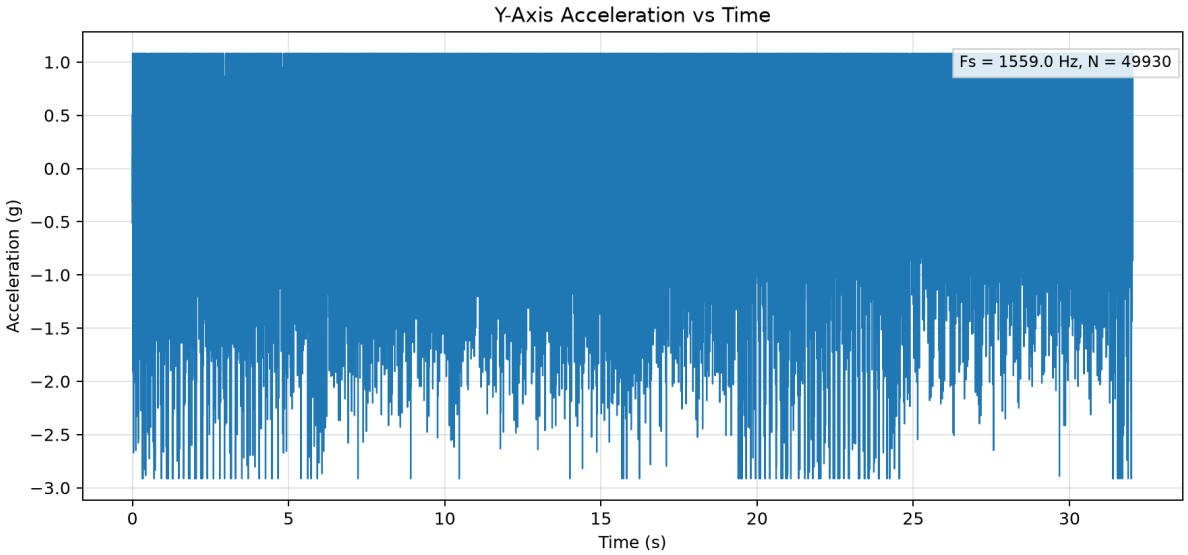



Figure 2. Y-axis calibrated acceleration versus time.

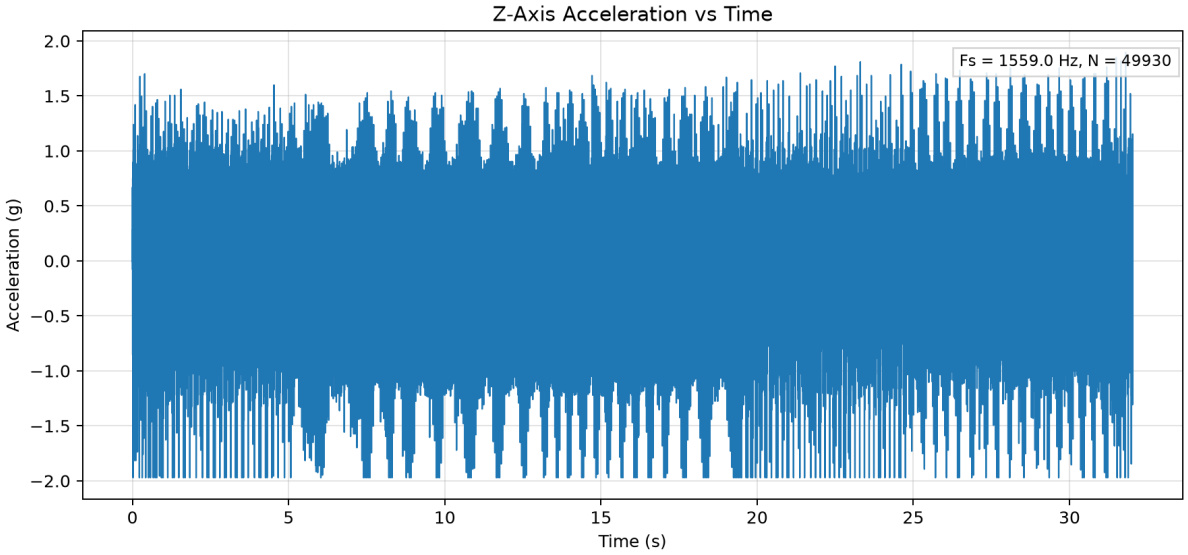



Figure 3. Z-axis calibrated acceleration versus time.

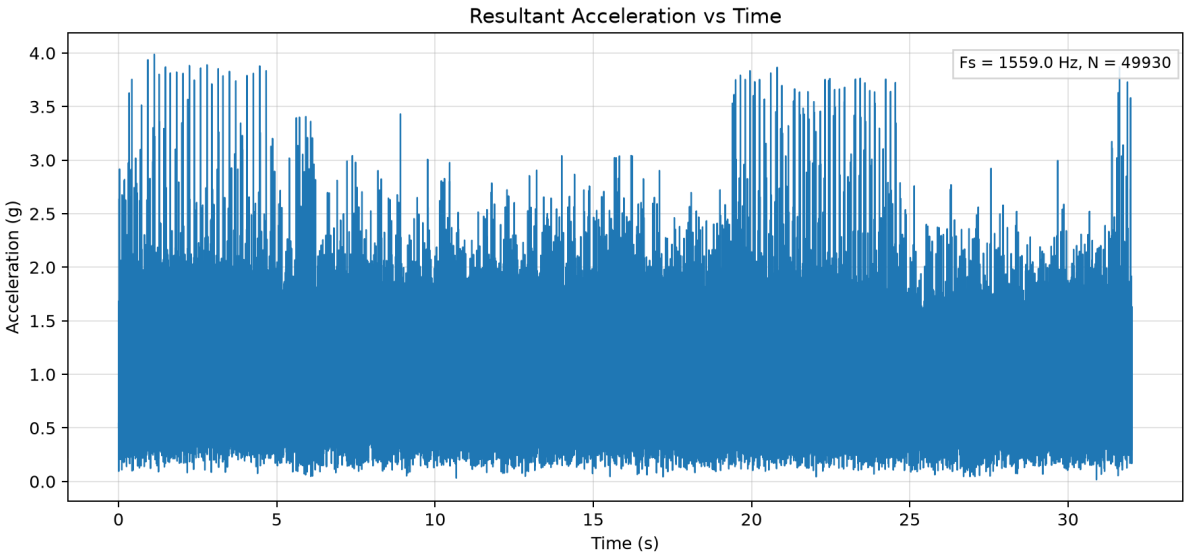



Figure 4. Resultant calibrated acceleration magnitude versus time.

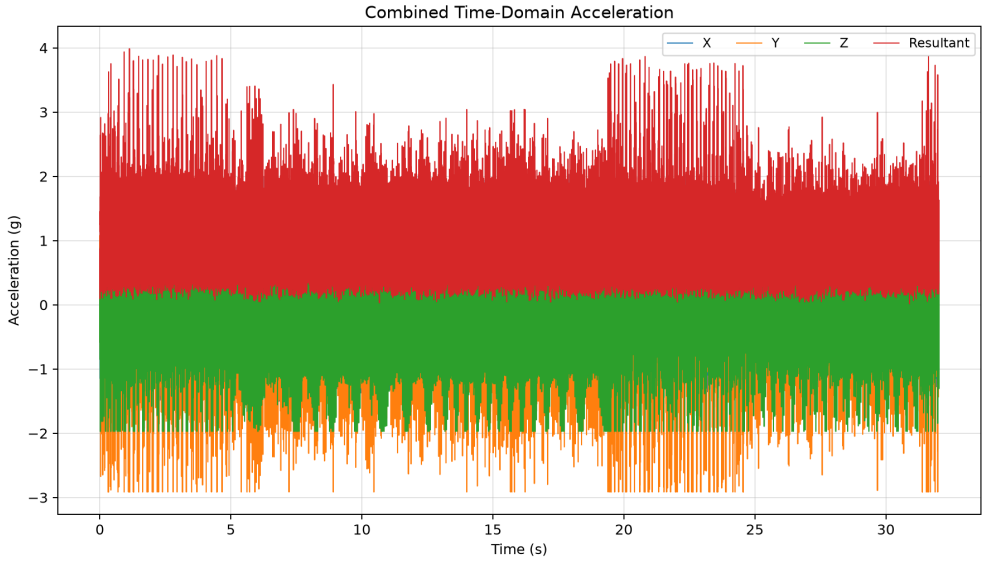



Figure 5. Combined time-domain acceleration comparison.

Table 1. Time-domain vibration features calculated from DC-removed AC signals.

| Feature | X | Y | Z | Resultant |
|---|---|---|---|---|
| Mean | 1.071e-17 | -6.262e-18 | 5.365e-17 | 4.497e-17 |
| Median | -0.02354 | 0.00586 | 0.04737 | -0.04963 |
| Standard Deviation | 0.57813 | 0.71111 | 0.55299 | 0.47399 |
| Variance | 0.33423 | 0.50568 | 0.30580 | 0.22466 |
| RMS | 0.57812 | 0.71110 | 0.55299 | 0.47398 |
| Maximum | 2.03506 | 1.14258 | 1.89893 | 3.02559 |
| Minimum | -1.87510 | -2.85351 | -1.96825 | -0.94479 |
| Peak | 2.03506 | 2.85351 | 1.96825 | 3.02559 |
| Absolute Peak | 2.03506 | 2.85351 | 1.96825 | 3.02559 |
| Peak-to-Peak | 3.91016 | 3.99609 | 3.86719 | 3.97038 |
| Range | 3.91016 | 3.99609 | 3.86719 | 3.97038 |
| Crest Factor | 3.52012 | 4.01280 | 3.55932 | 6.38335 |
| Shape Factor | 1.28425 | 1.30852 | 1.28224 | 1.30112 |
| Impulse Factor | 4.52070 | 5.25080 | 4.56390 | 8.30549 |
| Clearance Factor | 5.41007 | 6.40223 | 5.48551 | 9.81317 |
| Skewness | 0.30781 | -0.52344 | -0.47875 | 1.11798 |
| Kurtosis | 0.44415 | 0.81082 | 0.44784 | 2.98541 |
| Mean Absolute Value | 0.45016 | 0.54344 | 0.43127 | 0.36429 |
| Energy | 1.669e+04 | 2.525e+04 | 1.527e+04 | 1.122e+04 |
| Signal Power | 0.33422 | 0.50567 | 0.30579 | 0.22466 |
| Median Absolute Deviation | 0.35938 | 0.41406 | 0.35156 | 0.29787 |
| Q1 | -0.36729 | -0.38476 | -0.33544 | -0.33867 |
| Q2 / Median | -0.02354 | 0.00586 | 0.04737 | -0.04963 |
| Q3 | 0.35146 | 0.44336 | 0.37550 | 0.25584 |
| Interquartile Range | 0.71875 | 0.82812 | 0.71094 | 0.59451 |
| Zero-Crossing Rate | 0.35020 | 0.31479 | 0.27978 | 0.36508 |

# 5. Time-Domain Feature Definitions

| Feature | Definition | Engineering Meaning |
|---|---|---|
| Mean | mean(x) | Average acceleration level. For AC vibration it should be close to zero after DC removal. |
| Median | middle value of x | Robust central value that is less sensitive to isolated peaks than the mean. |
| Standard Deviation | sqrt(sample variance) | Spread of vibration amplitude around the mean; higher values indicate greater variation. |
| Variance | sum((x-mean)^2)/(N-1) | Mean-square spread of the signal; useful for comparing vibration intensity. |
| RMS | sqrt(mean(x^2)) | Overall vibration intensity or energy level of the acceleration signal. |
| Peak | max(abs(x)) | Largest absolute acceleration excursion recorded in the signal. |
| Peak-to-Peak | max(x)-min(x) | Total excursion range between positive and negative extremes. |
| Crest Factor | peak/RMS | Peak severity relative to average vibration level; higher values may indicate impulsive behavior. |
| Shape Factor | RMS/mean(abs(x)) | Describes waveform shape independent of absolute amplitude. |
| Impulse Factor | peak/mean(abs(x)) | Highlights isolated high-amplitude events relative to average absolute motion. |
| Clearance Factor | peak/(mean(sqrt(abs(x)))^2) | Sensitive to sharp peaks and often used when inspecting impulsive vibration characteristics. |
| Skewness | third standardized moment | Measures asymmetry of amplitude distribution. |
| Kurtosis | excess fourth standardized moment | Measures heaviness of tails and peakedness; larger values can be associated with impulsive events. |
| Energy | sum(x^2) | Total discrete signal energy over the acquisition duration. |
| Signal Power | mean(x^2) | Average energy per sample; equal to RMS squared. |
| Zero-Crossing Rate | fraction of sign changes | Indicates how often the AC signal crosses zero and can reflect frequency content. |

## 6. Frequency-Domain Analysis

Single-sided amplitude spectra are computed from DC-removed signals. The frequency associated with an FFT bin is a spectral component frequency, not automatically the motor shaft speed.

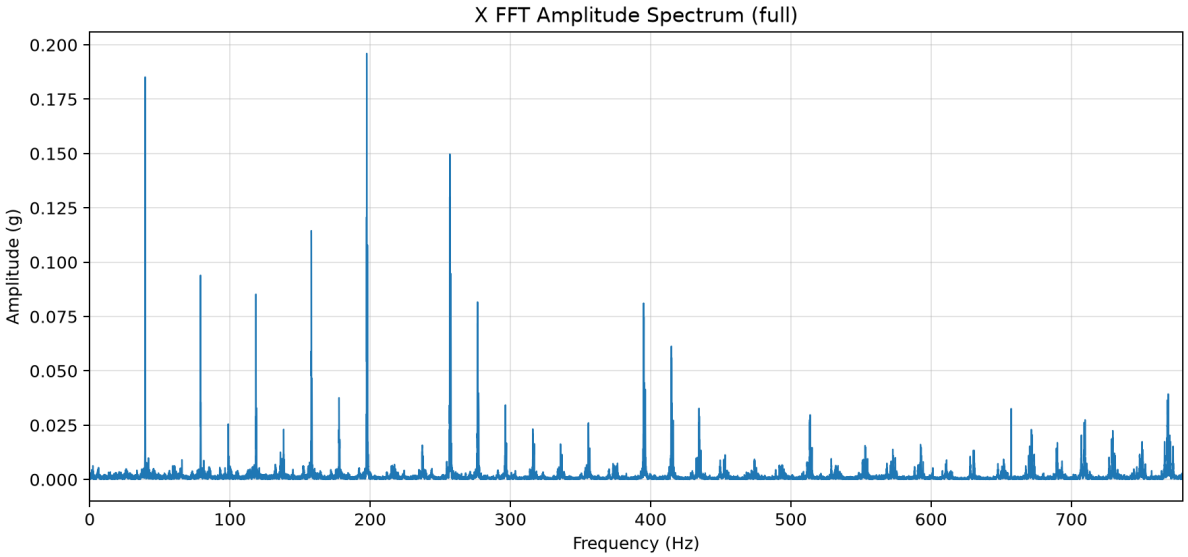



Figure 6. X-axis full FFT spectrum.

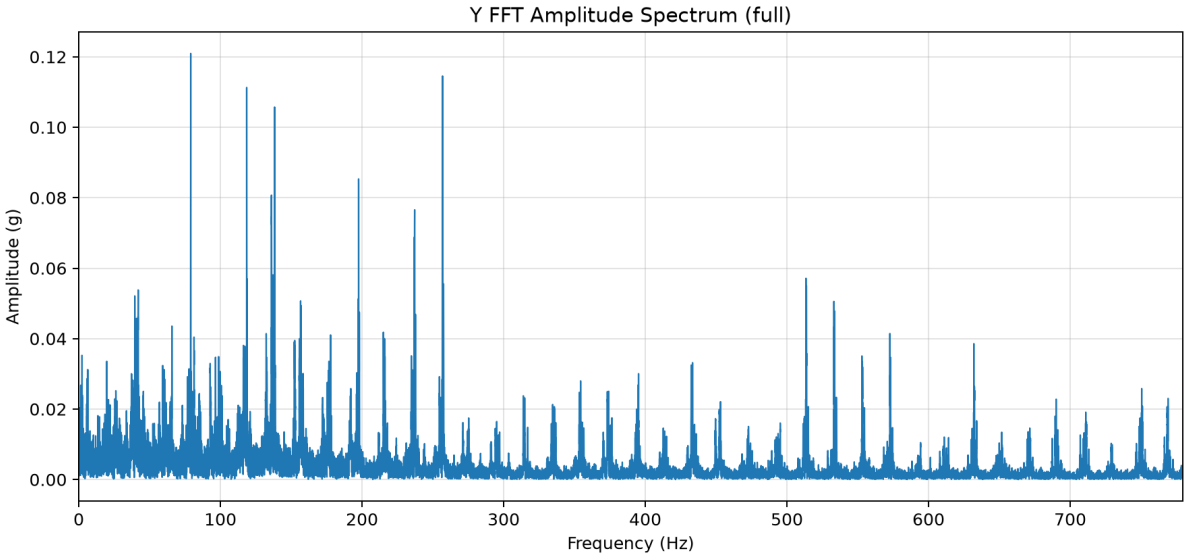



Figure 7. Y-axis full FFT spectrum.

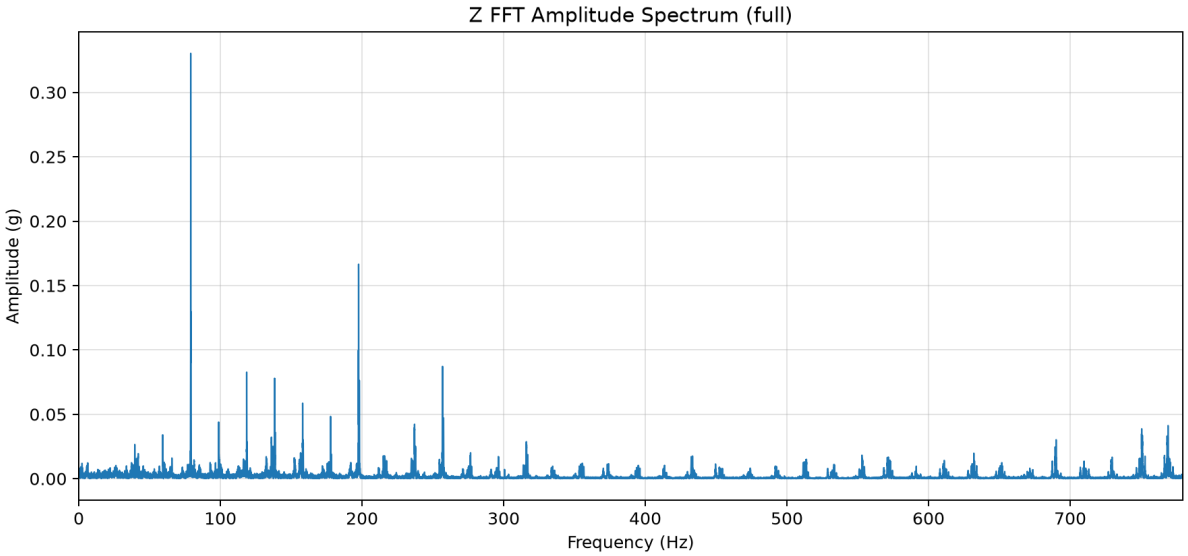



Figure 8. Z-axis full FFT spectrum.

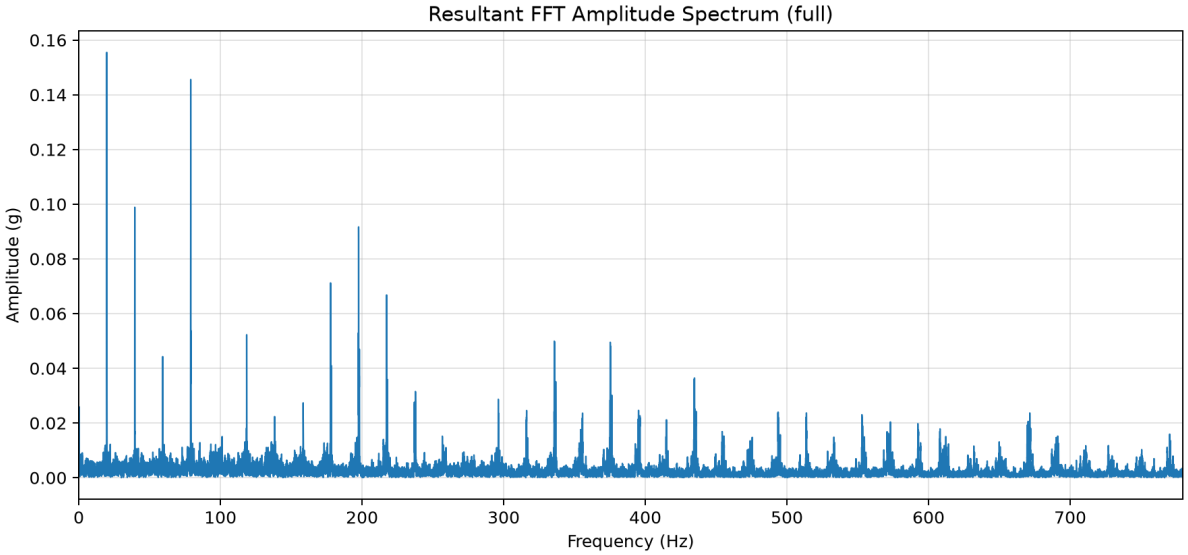



Figure 9. Resultant full FFT spectrum.

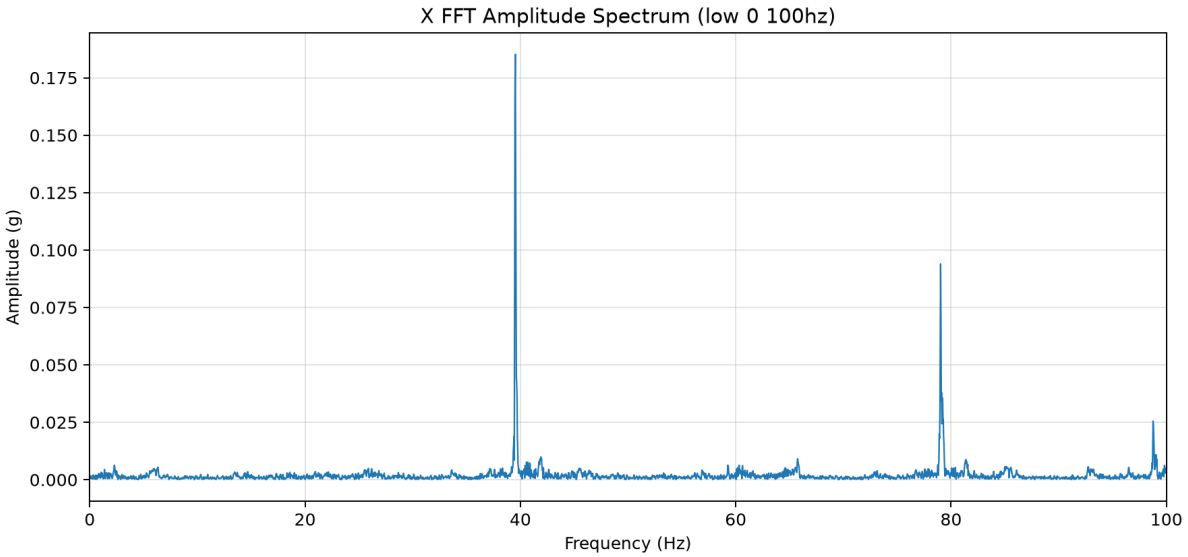



Figure 10. X-axis low-frequency FFT spectrum, 0-100 Hz.

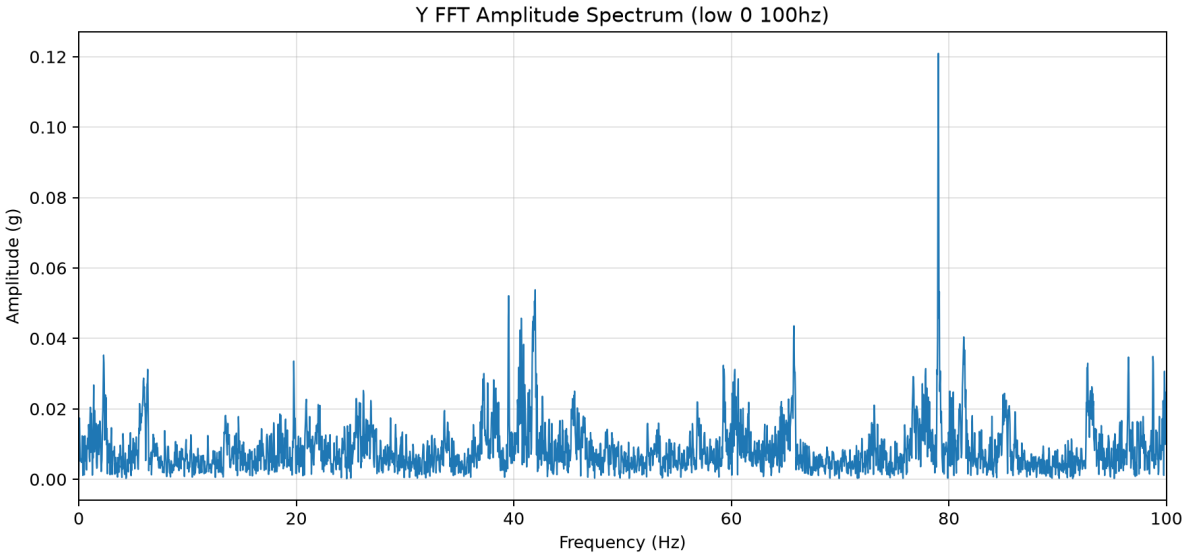



Figure 11. Y-axis low-frequency FFT spectrum, 0-100 Hz.

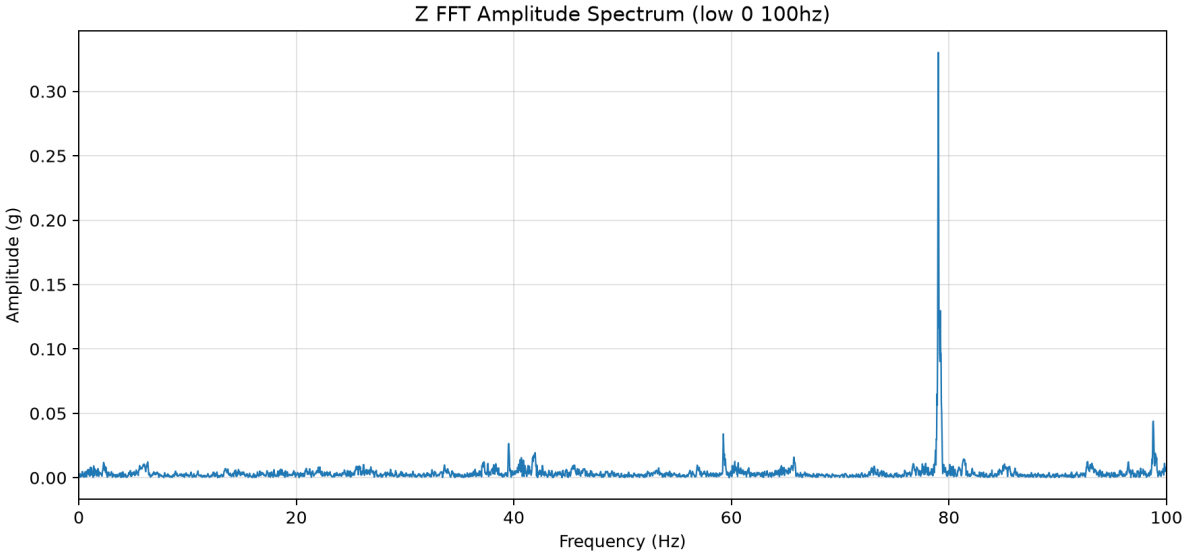



Figure 12. Z-axis low-frequency FFT spectrum, 0-100 Hz.

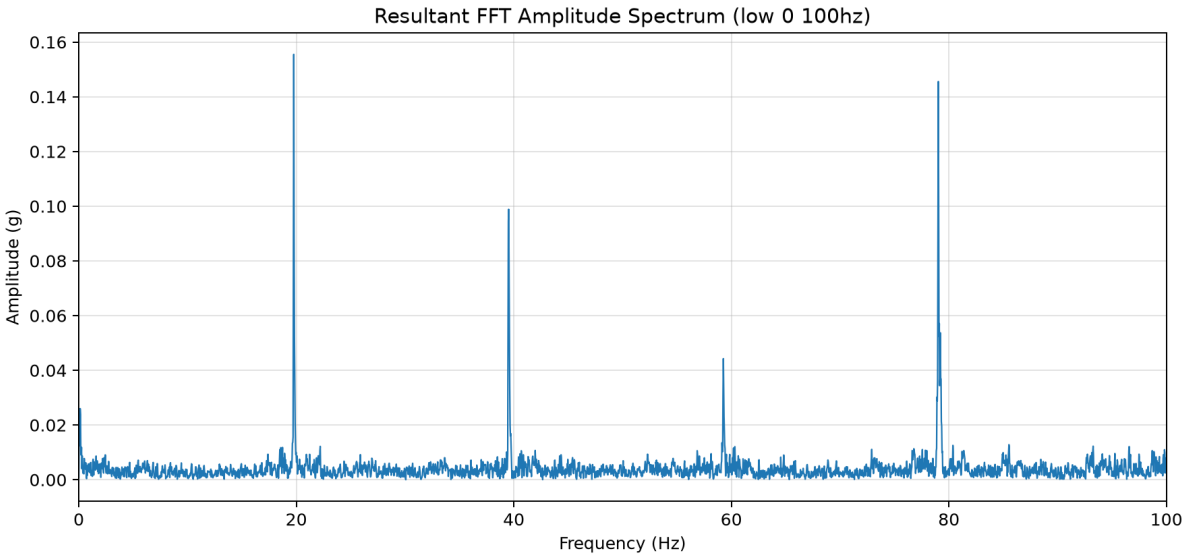



Figure 13. Resultant low-frequency FFT spectrum, 0-100 Hz.

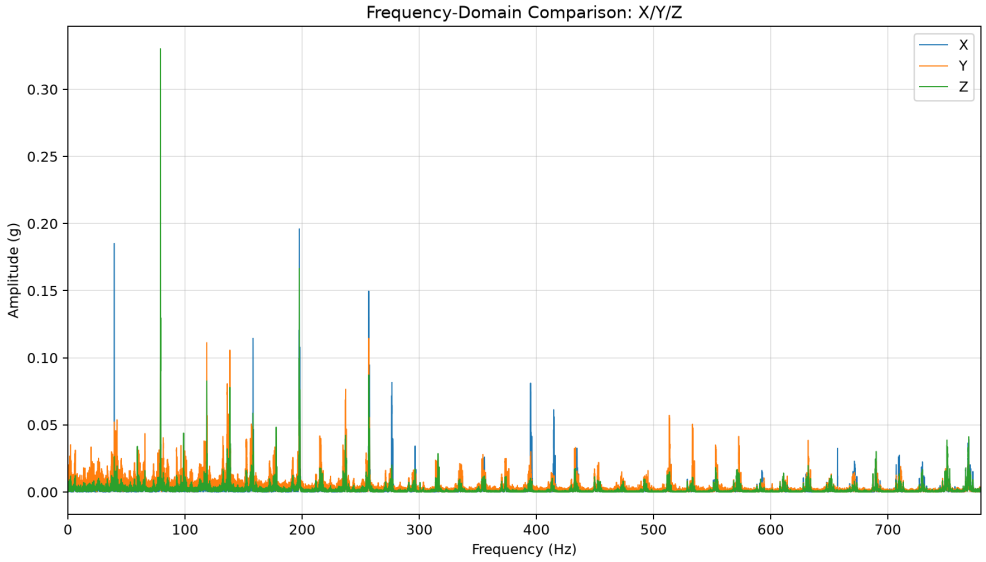



Figure 14. Full-spectrum comparison of X/Y/Z axes.

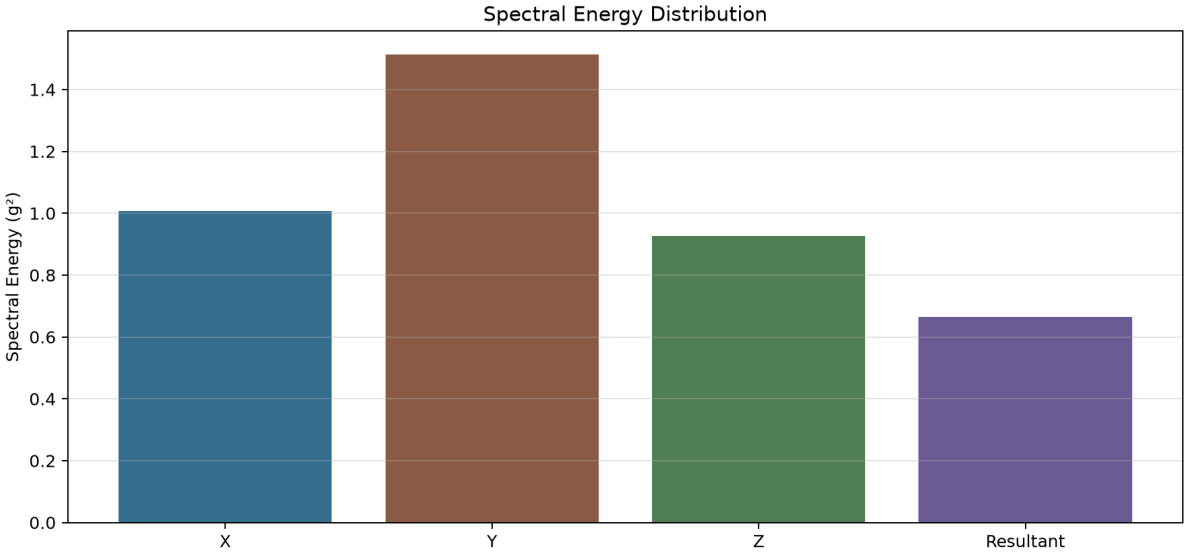



Figure 15. Spectral energy distribution comparison.

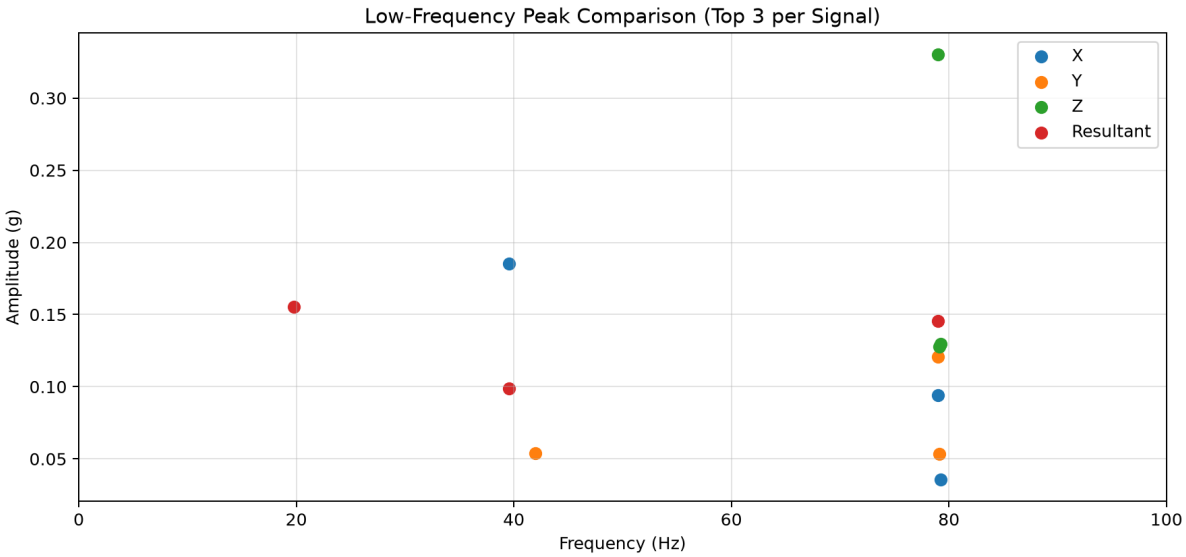



Figure 16. Low-frequency peak comparison.

# 7. Frequency-Domain Features

Table 2. Frequency-domain features calculated from Hann-windowed AC signals.

| Feature | X | Y | Z | Resultant |
|---|---|---|---|---|
| Dominant Frequency | 197.55243 | 79.02722 | 79.02722 | 19.76461 |
| Dominant Amplitude | 0.19601 | 0.12094 | 0.33026 | 0.15549 |
| Spectral Centroid | 367.20749 | 277.76898 | 310.38772 | 339.19280 |
| Spectral RMS Frequency | 318.18126 | 281.22976 | 279.02407 | 328.14325 |
| Spectral Bandwidth | 203.78211 | 189.05449 | 216.86082 | 218.36007 |
| Spectral Energy | 1.00681 | 1.51308 | 0.92684 | 0.66393 |
| Spectral Variance | 4.153e+04 | 3.574e+04 | 4.703e+04 | 4.768e+04 |
| Spectral Skewness | -0.41648 | 0.18151 | 0.02176 | -0.40625 |
| Spectral Kurtosis | 2.00031 | 2.26763 | 1.95334 | 1.74636 |
| Spectral Flatness | 0.49950 | 0.57010 | 0.48524 | 0.68529 |
| Spectral Entropy | 0.57351 | 0.79206 | 0.57782 | 0.74563 |
| Spectral Roll-off | 414.77581 | 433.32269 | 296.34426 | 509.28999 |
| Spectral Peak | 0.19601 | 0.12094 | 0.33026 | 0.15549 |
| Number of Significant Peaks | 216.00000 | 1618.00000 | 130.00000 | 684.00000 |

| Feature | Definition | Engineering Meaning |
|---|---|---|
| Dominant Frequency | frequency of largest non-DC FFT amplitude | Strongest periodic component in the measured vibration spectrum. |
| Dominant Amplitude | largest non-DC FFT amplitude | Amplitude of the strongest spectral component in g. |
| Spectral Centroid | sum(f*A)/sum(A) | Amplitude-weighted center frequency; higher values indicate energy shifted toward higher frequencies. |
| Spectral RMS Frequency | sqrt(sum(f^2*P)/sum(P)) | Power-weighted frequency magnitude of the spectrum. |
| Spectral Bandwidth | sqrt(sum((f-centroid)^2*P)/sum(P)) | Width of spectral energy around the centroid. |
| Spectral Energy | sum(A^2) | Total spectrum energy measure from FFT amplitudes. |
| Spectral Flatness | geometric mean(A)/arithmetic mean(A) | Indicates whether the spectrum is tone-like or broadly distributed. |
| Spectral Entropy | normalized entropy of spectral power | Measures how concentrated or spread out the spectral power is. |
| Spectral Roll-off | frequency below which 85% of power lies | Shows the frequency range containing most spectral energy. |

## 8. Spectral Peak Analysis

Equivalent RPM is calculated as frequency x 60 and is labeled only as the equivalent rotational speed if that frequency corresponds to 1x shaft rotational frequency. The script does not assert that any component is definitely shaft speed.

Table 3. Most significant full-spectrum peaks.

| Axis | Rank | Frequency (Hz) | Amplitude (g) | Equivalent RPM |
|---|---|---|---|---|
| X | 1.0000 | 197.5524 | 0.1960 | 1.185e+04 |
| X | 2.0000 | 39.5292 | 0.1852 | 2371.7533 |
| X | 3.0000 | 256.8463 | 0.1497 | 1.541e+04 |
| X | 4.0000 | 197.6461 | 0.1420 | 1.186e+04 |
| X | 5.0000 | 256.7214 | 0.1300 | 1.540e+04 |
| X | 6.0000 | 256.9087 | 0.1297 | 1.541e+04 |
| X | 7.0000 | 197.4588 | 0.1208 | 1.185e+04 |
| X | 8.0000 | 158.0232 | 0.1145 | 9481.3927 |
| X | 9.0000 | 198.0832 | 0.1079 | 1.188e+04 |
| X | 10.0000 | 257.5332 | 0.0947 | 1.545e+04 |
| Y | 1.0000 | 79.0272 | 0.1209 | 4741.6331 |
| Y | 2.0000 | 256.8463 | 0.1145 | 1.541e+04 |
| Y | 3.0000 | 118.5252 | 0.1113 | 7111.5129 |
| Y | 4.0000 | 138.2898 | 0.1057 | 8297.3895 |
| Y | 5.0000 | 256.9087 | 0.1040 | 1.541e+04 |
| Y | 6.0000 | 256.7214 | 0.0987 | 1.540e+04 |
| Y | 7.0000 | 197.5524 | 0.0853 | 1.185e+04 |
| Y | 8.0000 | 136.0105 | 0.0807 | 8160.6297 |
| Y | 9.0000 | 237.1441 | 0.0766 | 1.423e+04 |
| Y | 10.0000 | 236.9568 | 0.0688 | 1.422e+04 |
| Z | 1.0000 | 79.0272 | 0.3303 | 4741.6331 |
| Z | 2.0000 | 197.5524 | 0.1665 | 1.185e+04 |
| Z | 3.0000 | 79.2146 | 0.1297 | 4752.8736 |
| Z | 4.0000 | 79.1209 | 0.1276 | 4747.2534 |
| Z | 5.0000 | 197.4588 | 0.0998 | 1.185e+04 |
| Z | 6.0000 | 256.8463 | 0.0872 | 1.541e+04 |
| Z | 7.0000 | 118.5252 | 0.0827 | 7111.5129 |
| Z | 8.0000 | 138.2898 | 0.0780 | 8297.3895 |
| Z | 9.0000 | 256.7214 | 0.0772 | 1.540e+04 |
| Z | 10.0000 | 198.0832 | 0.0763 | 1.188e+04 |
| Resultant | 1.0000 | 19.7646 | 0.1555 | 1185.8766 |
| Resultant | 2.0000 | 79.0272 | 0.1456 | 4741.6331 |
| Resultant | 3.0000 | 39.5292 | 0.0989 | 2371.7533 |
| Resultant | 4.0000 | 197.5524 | 0.0917 | 1.185e+04 |
| Resultant | 5.0000 | 177.7878 | 0.0712 | 1.067e+04 |
| Resultant | 6.0000 | 217.3483 | 0.0668 | 1.304e+04 |

| Axis | Rank | Frequency (Hz) | Amplitude (g) | Equivalent RPM |
|---|---|---|---|---|
| Resultant | 7.0000 | 79.1209 | 0.0571 | 4747.2534 |
| Resultant | 8.0000 | 79.2146 | 0.0537 | 4752.8736 |
| Resultant | 9.0000 | 197.4588 | 0.0530 | 1.185e+04 |
| Resultant | 10.0000 | 118.5252 | 0.0523 | 7111.5129 |

Table 4. Most significant low-frequency peaks from 0-100 Hz.

| Axis | Rank | Frequency (Hz) | Amplitude (g) | Equivalent RPM |
|---|---|---|---|---|
| X | 1.0000 | 39.5292 | 0.1852 | 2371.7533 |
| X | 2.0000 | 79.0272 | 0.0940 | 4741.6331 |
| X | 3.0000 | 79.2146 | 0.0356 | 4752.8736 |
| X | 4.0000 | 98.7606 | 0.0255 | 5925.6363 |
| X | 5.0000 | 39.4043 | 0.0189 | 2364.2596 |
| X | 6.0000 | 99.0416 | 0.0108 | 5942.4971 |
| X | 7.0000 | 41.9022 | 0.0099 | 2514.1334 |
| X | 8.0000 | 65.7571 | 0.0091 | 3945.4284 |
| X | 9.0000 | 81.3690 | 0.0088 | 4882.1398 |
| X | 10.0000 | 40.5596 | 0.0077 | 2433.5762 |
| Y | 1.0000 | 79.0272 | 0.1209 | 4741.6331 |
| Y | 2.0000 | 41.9647 | 0.0538 | 2517.8802 |
| Y | 3.0000 | 79.1209 | 0.0533 | 4747.2534 |
| Y | 4.0000 | 39.5292 | 0.0521 | 2371.7533 |
| Y | 5.0000 | 41.9022 | 0.0506 | 2514.1334 |
| Y | 6.0000 | 41.8086 | 0.0462 | 2508.5131 |
| Y | 7.0000 | 40.6845 | 0.0457 | 2441.0699 |
| Y | 8.0000 | 41.7461 | 0.0449 | 2504.7663 |
| Y | 9.0000 | 65.7571 | 0.0436 | 3945.4284 |
| Y | 10.0000 | 40.5596 | 0.0423 | 2433.5762 |
| Z | 1.0000 | 79.0272 | 0.3303 | 4741.6331 |
| Z | 2.0000 | 79.2146 | 0.1297 | 4752.8736 |
| Z | 3.0000 | 79.1209 | 0.1276 | 4747.2534 |
| Z | 4.0000 | 98.7918 | 0.0439 | 5927.5097 |
| Z | 5.0000 | 59.2626 | 0.0340 | 3555.7565 |
| Z | 6.0000 | 39.5292 | 0.0265 | 2371.7533 |
| Z | 7.0000 | 78.7774 | 0.0215 | 4726.6457 |
| Z | 8.0000 | 41.9647 | 0.0195 | 2517.8802 |
| Z | 9.0000 | 65.7571 | 0.0160 | 3945.4284 |
| Z | 10.0000 | 40.6845 | 0.0156 | 2441.0699 |
| Resultant | 1.0000 | 19.7646 | 0.1555 | 1185.8766 |
| Resultant | 2.0000 | 79.0272 | 0.1456 | 4741.6331 |
| Resultant | 3.0000 | 39.5292 | 0.0989 | 2371.7533 |
| Resultant | 4.0000 | 79.1209 | 0.0571 | 4747.2534 |

| Axis | Rank | Frequency (Hz) | Amplitude (g) | Equivalent RPM |
|---|---|---|---|---|
| Resultant | 5.0000 | 79.2146 | 0.0537 | 4752.8736 |
| Resultant | 6.0000 | 59.2626 | 0.0443 | 3555.7565 |
| Resultant | 7.0000 | 0.1249 | 0.0260 | 7.4937 |
| Resultant | 8.0000 | 0.0624 | 0.0240 | 3.7468 |
| Resultant | 9.0000 | 85.5218 | 0.0128 | 5131.3050 |
| Resultant | 10.0000 | 80.3698 | 0.0126 | 4822.1903 |

## 9. Feature Interpretation

RMS represents the overall vibration energy/intensity of the measured acceleration signal. An increase generally indicates stronger vibration under the tested condition.

Peak and peak-to-peak values describe the largest acceleration excursions. Larger values may indicate stronger transient or periodic vibration, but they should be compared across controlled experiments.

Crest factor, impulse factor, clearance factor, and kurtosis emphasize impulsive behavior. Higher values can be associated with sharper events, but none of these features alone proves a fault.

Dominant frequency is the largest non-DC spectral component. It identifies the strongest periodic content in the measured signal.

Spectral centroid and RMS frequency indicate where spectral content is concentrated. Higher values generally mean energy has shifted toward higher frequencies.

Spectral bandwidth, entropy, and flatness describe how concentrated or spread out the spectrum is. These features are useful for comparing operating or physical conditions.

Low-frequency peaks in the 0-100 Hz range are separated because motor rotational and harmonic components often appear in this region, depending on the experimental setup.

## 10. Observed Vibration Characteristics

The highest RMS vibration level is observed in Y with RMS = 0.7111 g.

The largest absolute peak is observed in Resultant with peak = 3.0256 g.

The strongest dominant spectral amplitude is observed in Z: 79.0272 Hz at 0.3303 g.

The strongest component in the 0-100 Hz band is on Z at 79.0272 Hz with amplitude 0.3303 g.

The highest crest factor is on Resultant (6.3833), which may indicate stronger transient content and should be compared with other controlled conditions.

The highest kurtosis is on Resultant (2.9854), which provides an indication of distribution peakedness but is not a fault diagnosis by itself.

## 11. Conclusion

This report summarizes measured vibration characteristics from the provided calibrated ADXL345 dataset. The extracted features and plots provide a basis for comparing this condition against other experimentally controlled motor conditions. No automatic health/fault label is assigned.

# 12. Appendix

## Processing Parameters

| Parameter | Value |
|---|---|
| FFT window | Hann |
| Low-frequency band | 0-100.0 Hz |
| Top peaks per signal | 10.0000 |
| Peak prominence ratio | 0.0300 |
| Spectral roll-off percentage | 0.8500 |
| Input file | dataload1\processed\adxl345_5sec_calibrated.csv |
| Output results folder | dataload1\results |

Complete feature tables and peak tables are saved as CSV files in data/results.